INCH-POUND

A-A-59795

21 Feb 2007

COMMERCIAL ITEM DESCRIPTION

TRUCK, LIFT, FORK, CONTAINER HANDLER, DIESEL-ENGINE-DRIVEN, PNEUMATIC-TIRED, ROUGH TERAIN 50,000-POUND CAPACITY

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) covers the general requirements for a commercial, diesel-engine-driven, all-wheel-drive, pneumatic tired, rough terrain container handling forklift truck (RTCH). The RTCH will primarily be used to lift, transport, and stack International Standards Organization (ISO) containers and sea/land containers of 20 and 40 foot lengths on paved, unimproved, mud, and snow surfaces. When using forks in the bottom lift configuration, the vehicle shall have the capability to lift, transport, and stack loads up to 50,000 pounds at a 48-inch load center. When using a top handling attachment, the vehicle shall have the capability to lift, transport, and stack loads (not including attachment) up to 44,000 pounds at a 48-inch load center.

2. SALIENT CHARACTERISTICS

2.1 Safety. The forklift shall comply with ASME/ANSI B56.6, UL 558, and OSHA standards in effect at the time of manufacture. An operator's seatbelt conforming to SAE J 386, and restraint devices designed to ensure the operator's upper body remains entirely within the protection of the overhead guard in the event of tipover shall be provided.

2.2 Engine. The forklift shall be powered by a standard, commercial, liquid cooled, diesel engine. The engine shall operate on diesel fuel in accordance with A-A-52557 without detrimental effects on the engine, or its performance.

2.3 Fuel system. The forklift shall have all necessary filters, water separators, and components required for diesel fuel operation. The forklift shall have a fuel tank of sufficient capacity to allow at least eight hours of continuous operation.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to WR-ALC/GBZO-ENV, 380 Richard Ray Blvd, Robins AFB GA 31098.

AMSC N/A FSC 3930

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2.4 Starting system. The starter switch shall not activate the engine starter while the engine is running, nor when the engine is not running and the transmission is in any forward or reverse gear. The starter switch shall only operate when the transmission is in the “neutral” or “park” positions.

2.5 Transmission. A torque converter and a powershift type of transmission shall be furnished, with at least three speeds forward and three speeds reverse. The transmission shall provide for positive inching control, or declutch control, that permits lifting of rated load at maximum lifting speed while the transmission control is in either forward or reverse direction, without vehicle movement. This control shall also apply the service brakes. When specified (see 6.2(c), a transmission oil cooler shall be furnished.

2.6 Hydraulic system. A pressure relief protection device is required along with pump(s), cylinders, control valves, filter(s), reservoir, hoses, and all other components necessary to make a complete hydraulic system. The hydraulic system shall also be designed to prevent forward tilt in the event of system failure; tilt angle shall not deviate more than two degrees if failure occurs at any position during load handling operations. The hydraulic lift system shall be designed so that in the event of hydraulic hose or pump failure, the load shall not descend (freefall) at an uncontrolled rate; lowering speed shall not exceed eight (8) inches per second under any load condition. A manual override for the tilt function shall also be provided to allow for load removal.

2.7 Uprights and carriage. With no load on the forks, the seated operator with seatbelt engaged shall be able to see at least one fork tip at all lift heights and fork positions.

2.8 Forks and carriage. Fork tines shall have the following dimensions. The forks shall be mounted so that when fully lowered and forklift operating in reverse, there is no damage or degradation to the fork tines, their attachment, or any other component of the carriage, including fork positioner and side shift.

a. Length: 96 inches (maximum)
b. Width: 11.8 inches (maximum)
c. Thickness: 4.5 inches maximum, except bend in fork heel may exceed 4.5 inches
d. Fork tip thickness: 1 ¼ inch (maximum)
e. Fork taper (underside of forks): Taper from fork tip rearward shall be 30 inches (minimum).
f. Fork carrier assembly width shall not exceed the overall width of the forklift
g. Fork tines shall be 48 inches in length with an underside taper. Fork tip thickness shall not exceed 1 ¼ inches.

2.9 Fork positioner. A hydraulically powered fork positioner shall be furnished and controlled by the operator while seated.

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2.10 Side shift. Hydraulically powered side shift capability shall be furnished. It shall be controlled by the seated operator, and shall provide at least 12-inch side shift capability (each side of center or 24-inches total travel).

2.11 Load backrest. The load backrest, in conjunction with the forks and carriage, shall provide a vertical rear load guard of at least 102.5 inches high, measured from the load carrying surfaces of the forks.

2.12 Steering. Power steering, with emergency steering in the event of power failure, shall be furnished.

2.13 Service brakes. Power brakes conforming to ASME/ANSI B56.6 shall be furnished, except maximum brake pedal force during stopping shall not exceed 80 pounds.

2.14 Parking brake. Parking brake shall conform to ASME/ANSI B56.6

2.15 Electrical system. A 24 volt direct current (VDC) electrical system shall be furnished. The electrical system shall consist of all electrical components necessary for operation of the forklift. The forklift shall have sufficient electrical grounding to prevent static discharge.

2.15.1 Backup alarm. An audible backup alarm shall be furnished. The alarm shall automatically activate when the transmission selector is placed in reverse.

2.16 Instruments and controls. In addition to the instruments supplied on the standard commercial forklift, an hour meter shall be installed in the instrument panel. Except the hour meter, all instruments shall illuminate. All load motion controls shall be right hand operation controls and of the self-centering type, i.e., controls shall return to the neutral position when released.

2.17 Lighting. Minimum of eight (8) sealed beam floodlights, six (6) in front facing forward, two of which shall illuminate the two near twist locks for all lift heights or shall be adjustable from the operators position, and two mounted in the rear facing rearward, shall be furnished. An additional floodlight shall be furnished that is adjustable from inside the cab by the operator while seated; the floodlight shall provide at least 270° rotation capability. Tail light(s) and brake stop light(s) shall be installed. All lights shall be protected by location or guards. Individual operator controlled switches shall be provided for front floodlights, rear floodlights, and adjustable lights.

2.18 Horn. A horn shall be furnished, with the horn actuator located within reach of the seated operator.

2.19 Operator's overhead guard. The guard shall be in accordance with ASME/ANSI B56.6. It shall also be a falling-object-protective-structure (FOPS) conforming to ASME/ANSI B56.6.

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2.20 Cab. The forklift shall be equipped with an enclosed cab that shall incorporate overhead protection requirements of ASME/ANSI B56.6. The cab shall have a door on each side of the vehicle with at least one window per door which can be opened for ventilation; the windows shall be capable of being locked in either the open or closed positions. The doors shall be provided with hinges that can be serviced without door removal and shall allow adjustments for door fit. The doors shall be capable of being locked in either the opened or closed positions. In addition to the door windows, the cab shall be furnished with a front windshield, rear window, and side windows. The windshield and all windows shall be safety glass. The cab interior shall be furnished with water-resistant non-absorbent thermal insulation. A commercial fire extinguisher shall be mounted on the truck in an easily accessible location. The fire extinguisher shall be a minimum 2-1/2 pound capacity ABC type, or equivalent. The fire extinguisher shall be UL listed.

2.21 Windshield wipers. There shall be at least two wipers furnished, one on the front windshield, and one on the rear window.

2.22 Winterization kits. There shall be Type A (-65°F) and Type C (-25°F) options available for winter protection.

2.22.1 Type A winterization. When specified (see 6.2(b), the truck shall be furnished with Type A winterization that shall protect to -65° F and shall consist of the following:

a. Power plant heaters. Engine coolant, engine oil and battery heaters shall be provided. All heaters shall operate on 110 volt, alternating current. A three wire, 25 feet long weatherproof cable of adequate capacity for all heaters being used simultaneously shall be provided. A stowage place shall be provided on the truck to keep the cable when not in use. The heaters shall be as follows:

    (1) The coolant heater shall be installed in the engine block or lower coolant inlet hose. A coolant-circulating pump, driven by a 110-volt alternating current motor shall be provided when a coolant inlet hose heater is furnished. The heater shall have adequate capacity to maintain engine coolant at a temperature of 10° F in an ambient temperature of -65° F. It shall be controlled to limit engine coolant to not more than 150° F.
    (2) An engine oil heater with adequate capacity to maintain engine oil at a temperature of at least +10 degrees F in an ambient temperature of -65° F shall be furnished. It shall be controlled to limit engine oil temperature to not more than 150 ° F.
    (3) A battery heater shall be provided. It shall have adequate capacity to maintain battery electrolyte at a temperature of at least +10° F in an ambient temperature of -65°F, and shall have a thermostat to limit temperature of the electrolyte to not more than +80°F.

b. Cab. The cab shall meet the requirements of 3.20, and shall also contain thermal insulation, if necessary, to meet the heating requirements of 3.23.1.c.
c. Heater and defroster. The heater shall be of sufficient capacity to maintain a temperature of +40° F at cab floor level in an ambient temperature of -40° F.

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d. Cold starting aid. A measured shot ether system or glow plug(s) shall be provided to assist in engine cold temperature starting.
e. Antifreeze. Engine coolant shall be protected to -65° F.

2.22.2 Type C winterization. When specified (see 6.2(b), the truck shall be furnished with Type C winterization that shall protect to -25° F and shall consist of the following:

a. Cold starting aid. Glow plug(s) or a measure shot ether injection system shall be furnished.
b. Cab. The cab shall meet the requirements of 3.20, and shall also contain thermal insulation, if necessary, to meet the heating requirements of 3.23.2.c.
c. Cab heater and defroster. The heater shall have sufficient capacity to maintain a temperature of at least +40° F at cab floor level in an ambient temperature of -20° F.
d. Antifreeze. Engine coolant shall be protected to -25° F with antifreeze.

2.23 Performance.

a. Capacity. Capability to safely handle a 50,000-pound load with a 48-inch load center, at all lift heights.
b. Lifting speed. At least 30 fpm.
c. Lowering speed. Not more than 80 fpm with rated load on forks and not more than 35 fpm unloaded.
d. Collapsed mast height. Upright height with forks on ground (collapsed mast height): 228 inches maximum, with no load on the forks.
e. Maximum forklift height. At least 143 inches with rated load on forks and forks horizontal 113.50 inches (with 20’ or 40’ handler kit)
f. Right angle turn. With rated load on the forks, forklift positioned perpendicular to a wall, and with the front of the load against the wall, the truck shall be able to back up and make a complete right angle turn (ending up parallel to the wall) within 210 inches.
g. Travel speed. 16 miles per hour (mph) minimum in both forward and reverse, while carrying rated load on the forks. The vehicle speed shall be governed so that maximum speed for safe operations is not exceeded.
h. Slope ascension. With the truck facing up the slope and with rated load on the forks, the truck shall be able to accelerate up a 25 percent slope from a dead stop.
i. Upright tilt. With no load on the truck, the forks shall have at least 5 of forward tilt, and at least 8 degrees of rear tilt.
j. Ground clearance. With rated load on the forks and mast vertical, ground clearance beneath the mast shall be at least 12 inches. With no load on the forks, the ground clearance of the truck other than the mast and axles shall be at least 7.0 inches.
k. Load drift. With hydraulic fluid at normal operating temperature, the lift assembly shall hold rated load at maximum lift height for at least 2.0 minutes with not more than 1.0 inch of vertical drift, and not more than 0.5 degree of rotational drift.
l. Stability. The forklift shall meet the ASME/ANSI B56.1 "forward stacking", "forward travel", "lateral stacking", and "lateral travel" stability requirements.

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m. Noise limits. Maximum allowable noise level shall not exceed 84dB at the operator's ear. If the noise level at the operator's ear exceeds 84 dB (85 to 92 dB) a permanent warning decal or placard shall be installed inside the cab clearly visible to the operator. Noise level limits that exceed 92dB are not acceptable.

2.24 Painting. The forklift shall be primed and painted using standard commercial practices. Unless otherwise specified (see 6.2(d). When specified the manufacturer must provide Forest Green (24052 of FED-STD-595) or Desert Sand (30313 of FED-STD-595).

2.25 Markings, data plates, and instruction plates. Corrosion resistant identification plate, and all warning and instruction plates shall be provided and mounted on the vehicle. The identification plate shall contain the following information:

Truck, Lift, Fork
Capacity: 50,000 pounds at (48) inch load center
Gross Vehicle Weight (GVW):
National Stock Number (NSN):
Contract Number:
Serial Number:
Model Number:
Registration Number:
Manufactured by:
Name:
Address:
Service Phone Number:
US Property:

2.25 Walkway coating. Floor plates and step surfaces shall be coated with a nonslip-coating compound, or be furnished with a nonslip metal or treadplate surface.

2.26 Slinging and tie down provisions. Permanently affixed slinging and tie down provisions that enable the truck to be lifted completely assembled and in its normal operational mode, and to be fastened to the floor or deck of a transportation medium, shall be provided. Provisions shall conform to MIL-STD-209, Classes1 and 2 or Class 3 for Type II equipment.

2.27 Electromagnetic interference (EMI). The forklift shall comply with EMI requirements of SAE J551-1 and SAE J551-2.

3. REGULATORY REQUIREMENTS.

3.1 The contractor is encouraged to use recovered materials to the maximum extent possible. However, used, rebuilt or remanufactured components, pieces, and parts shall not be incorporated in the forklift. The forklift shall comply with applicable Environmental Protection Agency (EPA) regulations.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Product Conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The RTCH will be in compliance with ISO 9001:2000. The government reserves the right to require proof of such conformance.

5. PACKAGING

5.1 Preservation, Packaging, Labeling and Marking. Unless otherwise specified, the preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc. This involves shipment from the supply source to the receiving activity, and reshipment from the receiving activity. The preservation, packaging, and packing shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities, labeling and marking shall be as specified in the contract and/or order.

6 NOTES

6.1 Source of documents.

6.1.1 Government documents. Copies of these documents are available online at www.assist.daps.dla.mil/quicksearch/or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

6.1.2 ASME /ANSI documents may be obtained at www.asme.org or from American Society of Mechanical Engineers, P.O. Box 2900 Fairfield NJ 07007.

6.1.3 ANSI documents may be obtained at www.ansi.org or from American National Standards Institute, 25 West 43rd Street, New York NY 10036.

6.1.4 SAE documents may be obtained at www.sae.org or from Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale PA 15096.

6.1.5 Underwriters Laboratories (UL) UL documents may be obtained at http://ulstandardsinfonet.ul.com/catalog or hardcopies and CD's can be found at COMM 2000 1414 Brook Drive Downers Grove, IL 60515

6.1.6 Copies for Occupational Safety and Health Association (OSHA) should be sent to the U. S. Department of Labor, 200 Avenue NW, Room 423, Washington DC 20210

6.1.7 ISO standards may be obtained at http://www.iso.org/iso/en/ISOOnline.frontpage or hardcopies may be found at International Organization for Standardization (ISO) 1, rue de Varembé, Case postale 56 CH-1211 Geneva 20, Switzerland

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6.2 Ordering data. Acquisition documents must specify the following:

a. Title, CID number, and date of this document.
b. Winterization (type A or Type C) if required (see 2.22)
c. Transmission oil cooler, if required (see 2.5).
d. Fork Lift primed and painted, if different (see 2.24).
e. Packaging requirements, if different (see 5.1).

6.3 Subject term (key word) listing.

1. Forklift
2. Rough Terrain Container Handler
3. RTCH
4. Rough Terrain Forklift
5. 50,000 Pound

MILITARY INTERESTS:

Custodian:
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

Preparing Activity:
Air Force - 84

Agent:
Air Force - 99

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